



Beijing-Dublin International College



SEMESTER II FINAL EXAMINATION – 2016/2017

Beijing-Dublin International College
BDIC1032J Maths 4 (Advanced Mathematics; Engineering)

HEAD OF SCHOOL: Zhongliang Liu
MODULE COORDINATOR: Yanru Ping

Time Allowed: 90 minutes

Instructions for Candidates

Answer ALL questions. The marks that each question carry is written as shown.

BJUT Student ID: _____ **UCD Student ID:** _____

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

Honesty Pledge: _____ **(Signature)**

Instructions for Invigilators

Non-programmable calculators are permitted. NO dictionaries are permitted.
No rough-work paper is to be provided for candidates.

SECTION A — GAP-FILLING QUESTIONS

This section is worth a total of **60** marks, with each question worth **4** marks.

1. Given that $f(x - y, x + y) = x^2 - y^2$, find the expression of $f(x, y) =$ _____.

2. Evaluate the limit

$$\lim_{(x,y) \rightarrow (0,2)} \frac{\sin xy}{x} = \text{_____}.$$

3. Determine the shape of the surface described by the following equation

$$x^2 + y^2 = 3z.$$

(a) An ellipsoid ;

(b) A paraboloid ;

(c) A cone ;

(d) A cylindrical surface .

The answer is _____

4. Given that $x^2 + y^2 - x = 6z$, find the total differential

$$dz|_{(0,1)} = \text{_____}$$

5. A function $z = z(x, y)$ is determined by the following equation $z = e^{2x-3z} + 2y$. Compute

$$3 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \text{_____}$$

BDIC1032J Advanced Mathematics (Module 4; Engineering)

6. Consider a function

$$z = f(x^2 - y^2, e^{xy}),$$

where $f(u, v)$ has continuous second-order partial derivatives. Then try to compute

$$\frac{\partial z}{\partial x} = \underline{\hspace{2cm}} \text{ and } \frac{\partial z}{\partial y} = \underline{\hspace{2cm}}.$$

7. Let $f(x, y) = \frac{e^x}{x - y}$. Then $f'_x + f'_y = \underline{\hspace{2cm}}$.

8. Let Σ be a plane in the 3-dimensional space,

$$\Sigma : \quad x - 2y + 2z = 9.$$

Let $P(1, 2, 3)$ be a point outside Σ . Then the distance between P and Σ is $\underline{\hspace{2cm}}$.

9. In order to find the extreme value of the the function $u = xy + 2yz$ with restriction condition $x^2 + y^2 + z^2 = 10$, we can appeal to the method of Lagrange Multiplier. Then the auxiliary function we need to construct is $L(x, y, z, \lambda) = \underline{\hspace{2cm}}$.

10. Let Σ be a surface described by the equation

$$x^2 + y^2 + z^2 = 3.$$

The equation of the tangent plane to Σ at the point $P(1, -1, 1)$ is given by $\underline{\hspace{2cm}}$.

11. Let C be a curve in space

$$C : \quad \begin{cases} x = t^4, \\ y = t^3, \\ z = t^2. \end{cases}$$

Find the equation of the tangent line to C at the point $(1, 1, 1)$: $\underline{\hspace{2cm}}$.

BDIC1032J Advanced Mathematics (Module 4; Engineering)

12. Here we have two statements:

- **Statement 1:** A function $f(x, y)$ is of two partial derivatives at the point (x_0, y_0) .
- **Statement 2:** $f(x, y)$ is differentiable at the point (x_0, y_0) .

Their relationship can be determined as: **Statement 1** is a _____ of **Statement 2** (sufficient but not necessary condition/sufficient and necessary condition/necessary but not sufficient condition/neither sufficient nor necessary condition).

13. Consider a function

$$z = x^2(2 + y^2) + y \ln y.$$

Let $P(x_0, y_0)$ be its extreme-value point. Compute the coordinates of this point P , and classify the extreme value to be a local minimum or local maximum value.

14. Evaluate the integral

$$\iint_{D: x^2+y^2 \leq 4} \left(x^3 \cos y - y^3 x^2 + 3\sqrt{4 - x^2 - y^2} \right) dx dy = \underline{\hspace{2cm}}$$

(Hint: Take benefit of the symmetry and geometrical meaning of a double integral.)

15. Consider a double integral

$$I = \iint_D xy dx dy, \quad \text{with } D = \{ (x, y) | 0 \leq x \leq 1; 0 \leq y \leq x \}.$$

It can be converted to the polar coordinates as

- (a) $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{\sec \theta} r \cos \theta \cdot r \sin \theta \cdot dr;$
- (b) $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{\csc \theta} r \cos \theta \cdot r \sin \theta \cdot r dr;$
- (c) $I = \int_0^{\frac{\pi}{4}} d\theta \int_0^{\csc \theta} r \cos \theta \cdot r \sin \theta \cdot dr;$
- (d) $I = \int_0^{\frac{\pi}{4}} d\theta \int_0^{\sec \theta} r \cos \theta \cdot r \sin \theta \cdot r dr.$

Which of the above answers is correct? _____

BDIC1032J Advanced Mathematics (Module 4; Engineering)

SECTION B — EXTENDED ANSWER QUESTIONS

Write your answers on the **Examination Book** provided.

This section is worth a total of **40** marks. The marks of each question are as shown.

- 16. (7 marks)** Compute the double integral

$$I = \iint_{D=\{(x,y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}} |x^2 + y^2 - 1| dx dy.$$

- 17. (7 marks)** Evaluate the following triple integral

$$I = \iiint_{\Omega} (z + x \sin z + y \cos z) dv,$$

where Ω is a region enclosed by the plane $z = 2$ and the cone $z = \sqrt{x^2 + y^2}$.

- 18. (7 marks)** Evaluate the following iterated integral

$$I = \int_0^1 dx \int_x^1 \cos y^2 dy.$$

- 19. (7 marks)** Express the following triple integral as an iterated integral in spherical coordinates

$$I = \iiint_{\Omega} (x + y + z) dv,$$

where Ω is a region bounded by the cone $z = \sqrt{x^2 + y^2}$ and the hemisphere $z = \sqrt{2 - x^2 - y^2}$.

BDIC1032J Advanced Mathematics (Module 4; Engineering)

20. (7 marks) Let Ω be a three dimensional solid body bounded by the surfaces

$$z = x^2 + y^2 \quad \text{and} \quad z = \sqrt{2 - (x^2 + y^2)}.$$

Compute the volume of Ω .

21. (5 marks) Consider a domain D in two dimensions, bounded by the lines

$$y = 1, \quad y = \sin x, \quad x = -\frac{\pi}{2}, \quad x = \frac{\pi}{2}.$$

Compute the following double integral

$$\iint_D (xy^5 - 1) dx dy.$$

(Hint: Take benefit of the symmetry and geometrical meaning of a double integral to simplify your calculation.)

Glossary

Auxiliary function	辅助函数
Cone	锥面
Convert	转换
Cylindrical coordinates	柱坐标
Cylindrical surface	柱面
Differentiable	可微分的
Double integral	二重积分
Ellipsoid	椭球面
Iterated integral	累次积分
Lagrange multiplier	拉格朗日乘子法
Paraboloid	抛物面
Partial derivative	偏导数
Polar coordinates	极坐标
Symmetry	对称性
Tangent line	切线
Tangent plane	切平面
Total differential	全微分
Triple integral	三重积分